

Maria Monica Stoica

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EDUCATION

Imperial College London Oct 2022 – Jun 2027
PhD Candidate, Computer Science — Centre for Doctoral Training in Safe and Trusted AI
Topic: Monitoring and Explainability of Deep Neural Networks

University of Oxford Oct 2021 – Sep 2022
MSc in Advanced Computer Science, Merit Google DeepMind Scholar

Harvard College Sep 2013 – May 2017
Bachelor of Arts in Computer Science GPA: 3.6/4.0 (Harvard), 4.7/5.0 (MIT)
Began at MIT (2013–2015), then transferred to Harvard to complete the degree.

RESEARCH INTERESTS

Out-of-distribution detection, robustness, and monitoring/explainability of deep neural networks for computer vision.

PUBLICATIONS & WORKSHOPS

1. **M. Stoica**, A. Hekal, A. Lomuscio. “A Robust Out-of-Distribution Detection Framework via Synergistic Smoothing.” *CVPR 2026 Findings*.
2. **M. Stoica**, F. Leofante, A. Lomuscio. “Out-of-Distribution Detection using Counterfactual Distance.” *IJCAI 2026 Workshop on Explainable AI (XAI)*.
3. C. Di Bonaventura, M. Nwachukwu, **M. Stoica**. “Wanted: Personalised Bias Warnings for Gender Bias in Language Models.” *ACL 2025 Workshop on Gender Bias in Natural Language Processing*.

HONORS & AWARDS

- UKRI Doctoral Studentship, Centre for Doctoral Training in Safe and Trusted AI (full stipend and fees).
- Google DeepMind Scholarship, University of Oxford.

EXPERIENCE

Citadel — Quantitative Research Intern, Model Risk Research May 2026 – Nov 2026

- Developing quantitative frameworks to validate, stress-test, and assess the risk of complex trading and pricing models.
- Designing independent benchmark models to strengthen firm-wide risk management and model governance.

Meta — PhD Intern, Systems & Infrastructure Software Engineering Jun 2025 – Sep 2025

- Built LLM Model Context Protocol tools that surface internal data-infrastructure knowledge to LLMs, helping engineers debug and query systems faster.

NatWest Markets — FX Options Quantitative Developer, Associate Sep 2019 – Jul 2021

- Designed pricing libraries, risk systems, and monitoring tools for FX Options trading desks, working with front- and middle-office users.

SKILLS

Languages: Python, C++
ML & Tools: PyTorch, NumPy, Git

LEADERSHIP

Outreach Officer, *Imperial Women & Non-Binary in Computing* (2023–25); Diversity & Inclusion Committee, *Safe and Trusted AI CDT* (2023–present).